

WAITING FOR WEALTH: INHERITANCES AND HOUSEHOLD SAVINGS

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MOTIVATION

- Inheritances account for 25-50% of U.S. household wealth (Feiveson & Sabelhaus (2018), Gale & Scholz (1994). Modigliani (1989), Kotlikoff & Summers (1981))
 - Most bequests happen at end of life
- Life-cycle model: foundational model of household finance and economics
- Model ignores inheritance receipt
 - Anticipated inheritances reduce savings
 - The size, probability, and timing of inheritances all matter
 - Most retirement planning software asks about inheritance expectations

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BIG QUESTIONS:

1. What features of reality does the model get wrong without inheritance?
 - Miss positive tail risk in income
 - Overstate savings motive
 - Wealth inequality: underestimated at middle age, overestimated for old
2. Implications of inheritances on structural parameters?
 - Ignoring inheritances $\Rightarrow$ overestimate discount factor β
3. Not today
 - Are inheritance expectations correct?
 - Implications for MPCs

METHOD

- We extend standard life-cycle savings model to include inheritances
- Households are endowed with a parent
- Parents' wealth evolves stochastically
- Households receive an inheritance when their parent dies
- Calibrate inheritance processes using SCF, PSID, HRS, SSA, ...

RELATED LITERATURE

1. Empirical literature on inheritance and wealth persistence: Kotlikoff & Summers (1981, JPE), Modigliani (1988, JEP), Gale & Scholz (1994, JEP), Charles & Hurst (2003, JPE), Gokhale, Kotlikoff, Sefton, & Weale (2001, JPubE), Piketty (2011, QJE), Boserup, Kopczuk & Kreiner (2016, AER), Elinder, Erixson, Waldenstrom (2018, JPubE), ...

This paper: Endogenous responses to inheritance expectations and receipt

2. Life-cycle models with bequests (parent's side): De Nardi (2004, REStud), De Nardi, French & Jones (2015, JPE), Lockwood (2018, AER), Heimer, Myrseth & Schoenle (2019, JF), Kvaerner (2022, RFS), ...

This paper: Models child's side

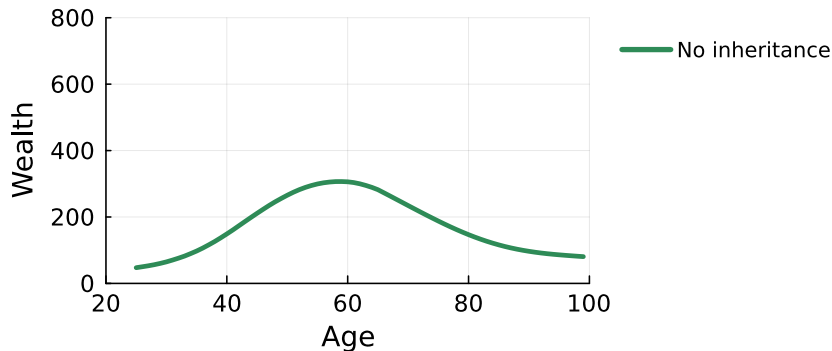
3. Anticipated inheritances and household behavior: Doorley & Pestel (2020, OBES), Coile & Weisbenner (2010, REStat), Kindermann, Mayr & Sachs (2020, JPubE), Druedahl & Martinello (2022, REStat) Aliprant, Carrol, & Young (2025), Asher (2026), ...

This paper: Structural model capturing probability, timing, and size jointly

- KMS (2020, JPubE) Close: estate taxes and labor response
- DM (2022, REStat) Close: unanticipated inheritances

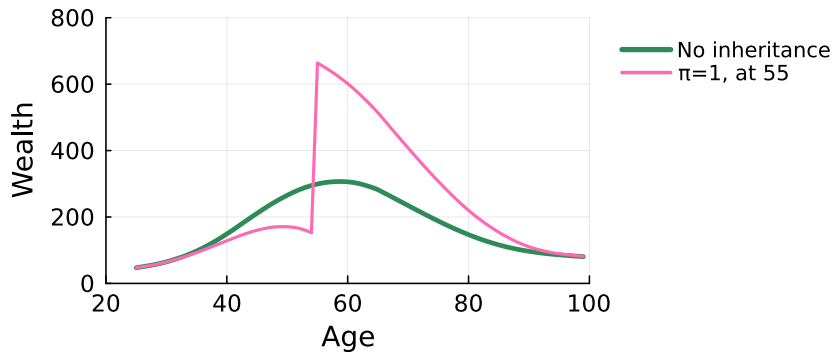
STYLIZED EXAMPLE

HOW DO LARGE INHERITANCES AFFECT AGGREGATE WEALTH?



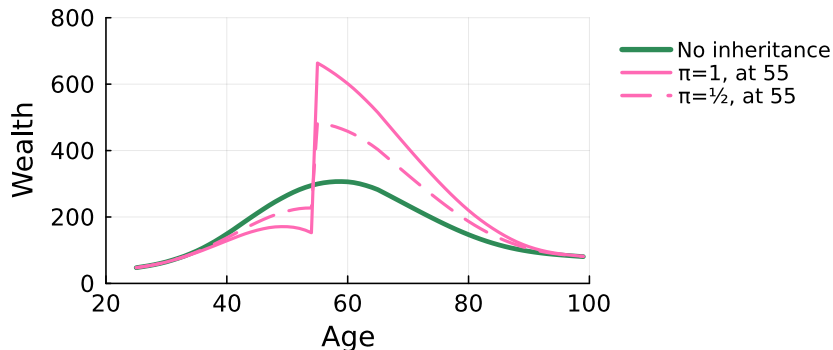
- **No inheritance**: standard hump-shaped wealth profile
- **Certain inheritance**: households **anticipate** inheritances and adjust savings
- 50% inheritance: **probability**, size, and timing of inheritance all matter
 - Uncertain timing

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A SIMPLE MODEL OF INHERITANCE RECEIPT

STANDARD LIFE-CYCLE MODEL

- Households live from age a 25 to 100, retire at 65
- CRRA preferences, warm-glow **bequest** motive
 - All papers have this; most ignore **inheritance receipt**
- Income y : deterministic life-cycle + persistent (v) + transitory (ε) shocks
- Save in a risk-free asset; no borrowing
- Budget constraint:

$$x' = (1 + r) (x - c + y(v, \varepsilon))$$

› Full model

ADDING INHERITANCES

- New state: parental status p — encodes survival and wealth
 - Dead (absorbing) or alive
 - Age-varying mortality π_a
 - Age-varying wealth quintiles $w_{q,a}$
 - Age-varying Markov chain for parental wealth: $w' \mid w, a \sim P_{ww'}(a)$
- Inheritance subject to taxes and fees $\tau(w)$
- Budget constraint:

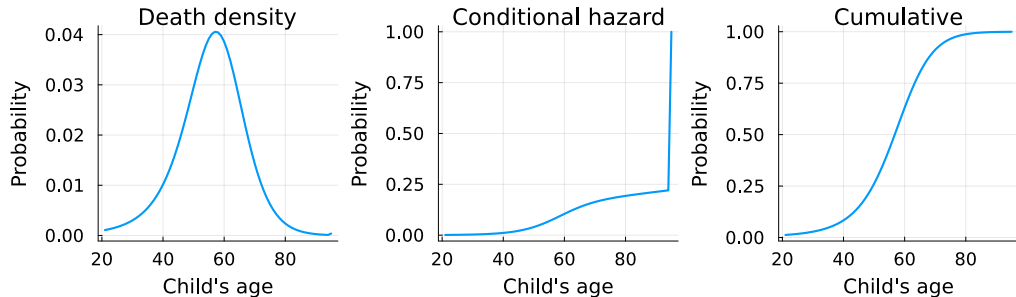
$$x' = (1 + r) (x - c + y(v', \varepsilon') + \mathbf{1}_{\text{death}} \cdot (1 - \tau(w)) w)$$

- Three sources of uncertainty: probability, size, and timing of inheritance

CALIBRATING THE PARENTAL PROCESSES

PARENTAL MORTALITY

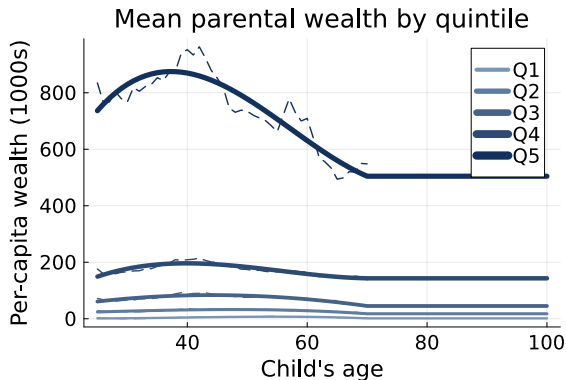
- Last parent death probabilities derived from SCF



Inheritance receipt at 50–70 $\Rightarrow$ expectations matter — but ignored by most models

PARENTAL WEALTH QUINTILES (HRS) - PER LIBEROS (CHILD)

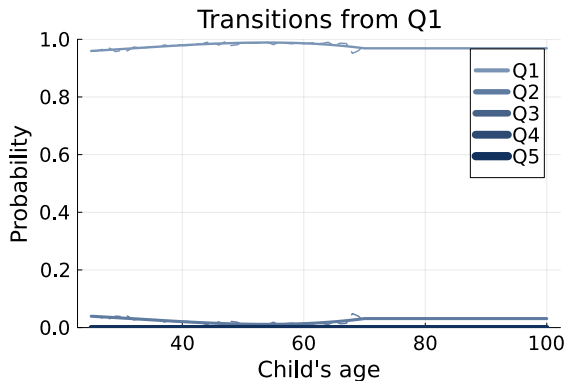
- Age-dependent quintile grid: $\mathcal{W}(a) = \{w_1(a), \dots, w_5(a)\}$



Top two quintiles: \$800k and \$200k $\Rightarrow$ relevant not only for “top 1 percent”

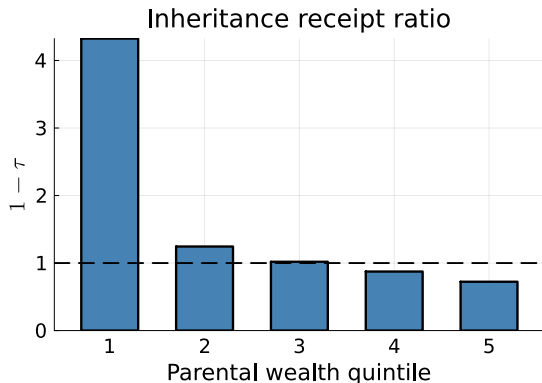
PARENTAL WEALTH TRANSITIONS (HRS)

- Markov chain with age-varying transition matrices: $\Pr(w' \mid w, a)$



INHERITANCE SLIPPAGE

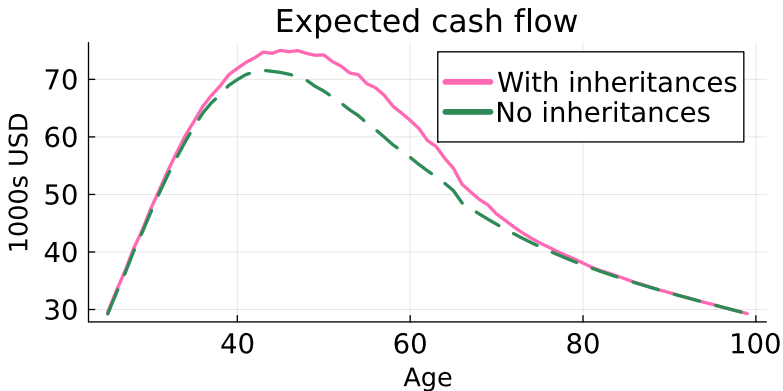
- Estimated from HRS exit interviews
- Slippage $\tau(w)$: fraction of parental wealth lost to taxes, fees, etc.



Poorest: bequests from life insurance, not wealth. Richest: ~30% lost to slippage.

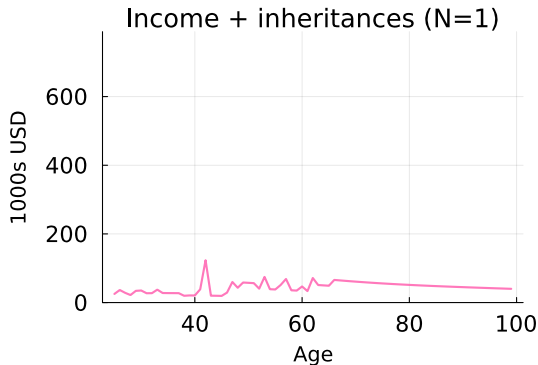
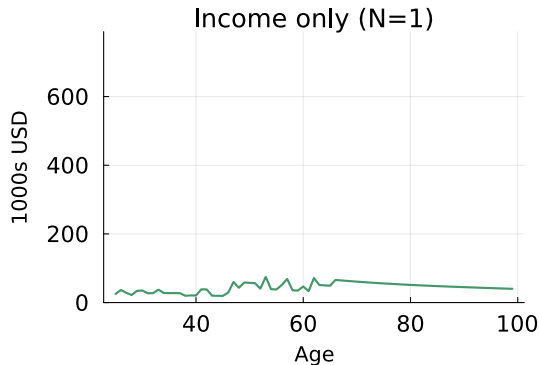
EFFECT OF INHERITANCES ON INCOME

EXPECTED INCOME: WITH VS. WITHOUT INHERITANCES



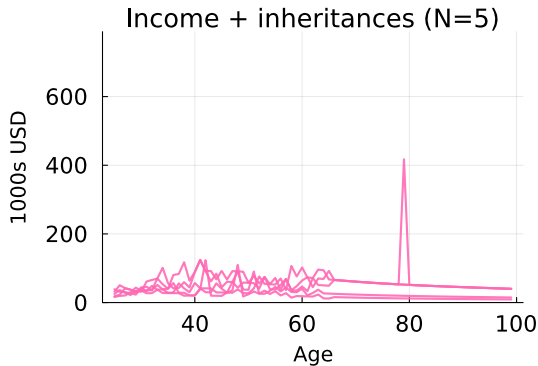
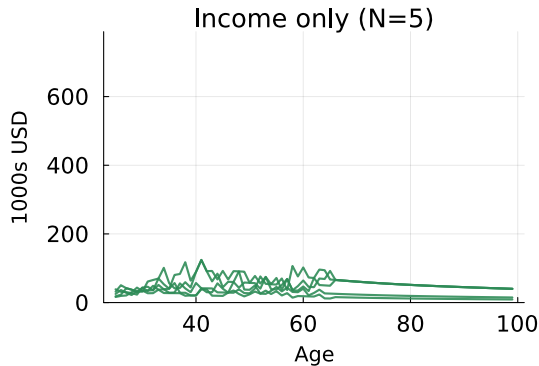
- Inheritances add a mid-life income bump

INHERITANCES GENERATES LARGE INCOME SPIKES



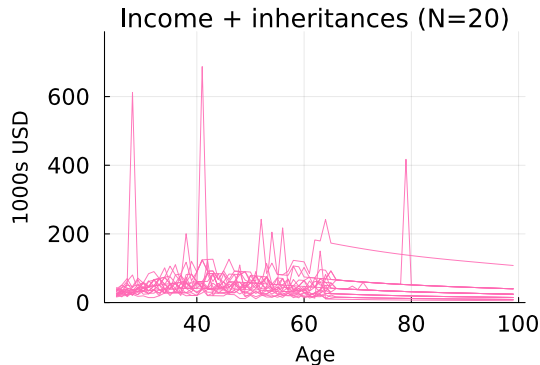
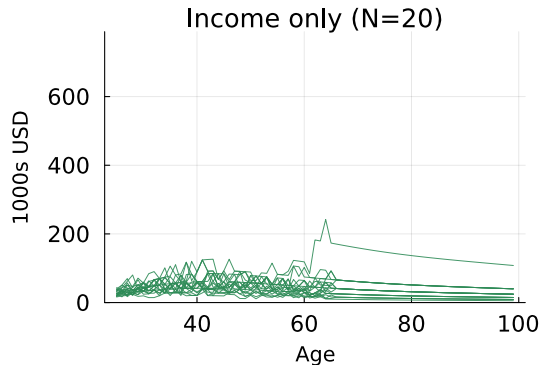
- Household #1: Spike in income at age 40
- Household #5: Huge inheritance at age 80
- ...Inheritances create large positive income spikes $\Rightarrow$ positive tail risk

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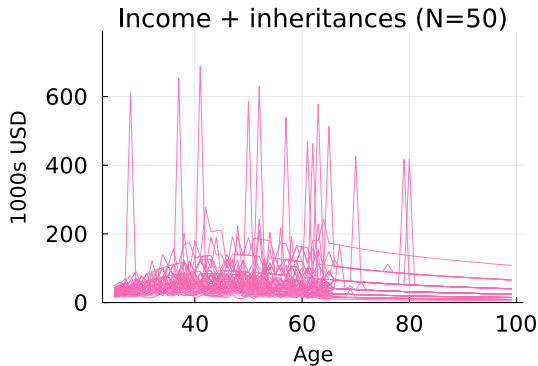
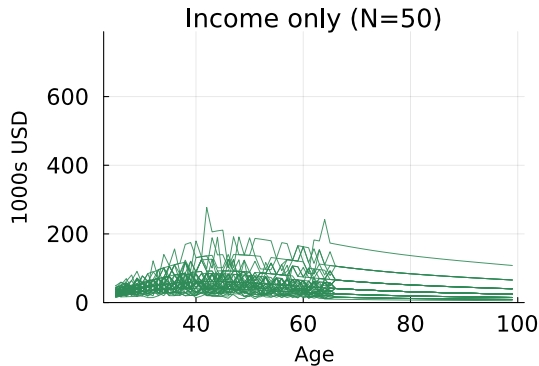
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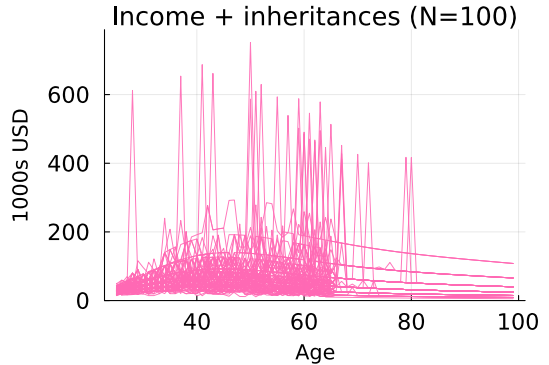
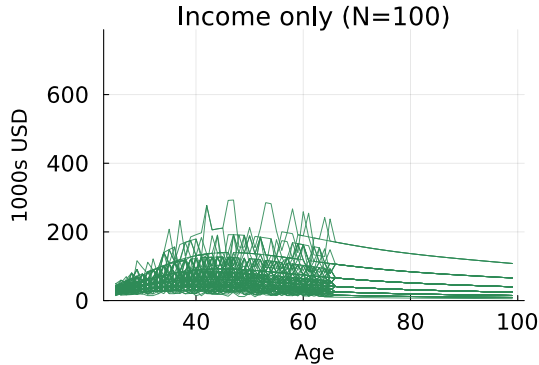
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ESTIMATION OF DISCOUNT FACTOR β

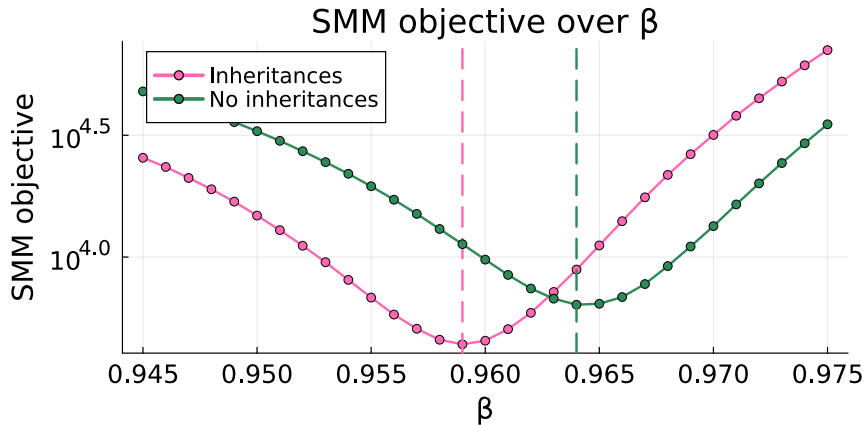
CALIBRATION

- We have now calibrated all inheritance processes
 - Age-dependent parental wealth quintiles
 - Age-dependent parental wealth transitions
 - Age-dependent parental mortality
 - $\Rightarrow$ Stochastic process determining inheritance receipt
- Estimate discount factor β
- Every else is standard $\Rightarrow$ take values from literature

IMPLICATIONS OF INHERITANCES ON SAVINGS CHOICES

- Inheritances $\Rightarrow$ higher life-time income $\Rightarrow$ need lower β
- Important question: How much?
- We estimate model with and without inheritances
 - Target wealth at age 40, 45, ...80

SMM OBJECTIVE FUNCTIONS



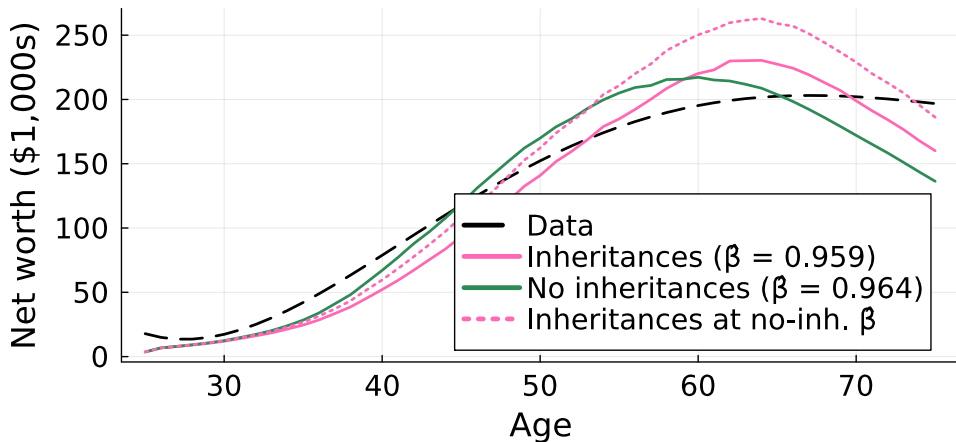
- Model with inheritances: lower β and better fit

DOES $\Delta\beta$ MATTER?

- $\hat{\beta} - \hat{\beta} = 0.05 \Rightarrow$ sounds small?
- Implies 50 basis points lower real risk-free rate
- Present value of \$100
 - 1 year: \$95.9 vs \$96.4 (0.5% difference)
 - 10 years: \$65.6 vs \$69.3 (5% difference)
 - 50 years: \$12.3 vs \$16.0 (23% difference)
- Life-cycle models have even longer horizons!

MODEL FIT

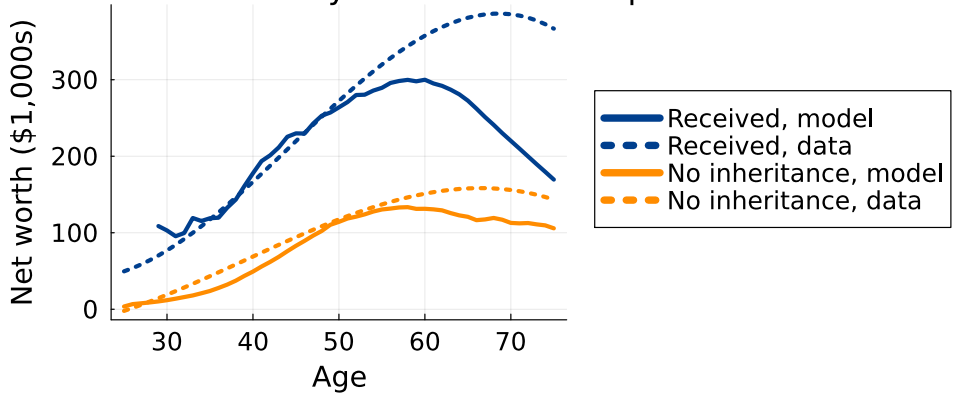
Model fit: median net worth



- Inheritances: Slower wealth accumulation and later peak wealth
- Both: too fast wealth decumulation in retirement

NET WORTH BY INHERITANCE RECEIPT

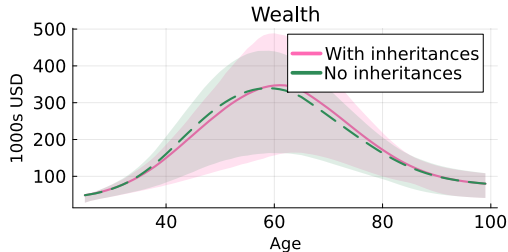
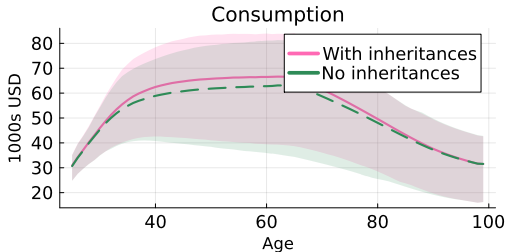
Median net worth by inheritance receipt



- Model matches non-targeted receipt gap well in working age
- HRS + model = SCF wealth gaps

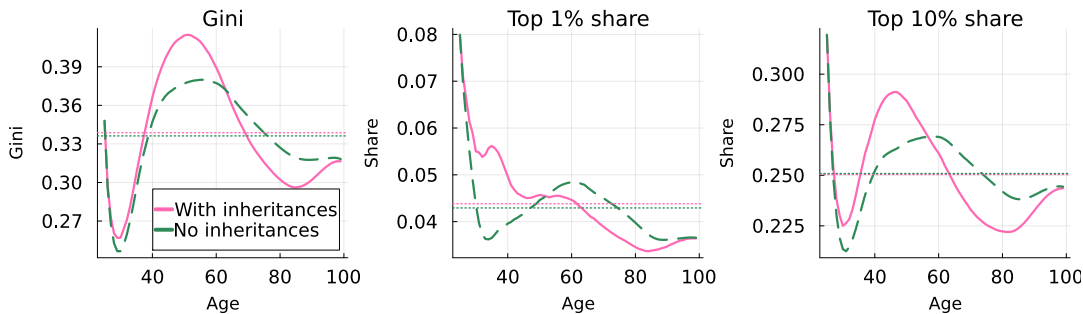
AGGREGATE EFFECTS OF INHERITANCES

LIFE-CYCLE PROFILES: WITH VS. WITHOUT INHERITANCES



- Inheritances increase consumption at all ages
- Inheritances increase wealth for older households
- Inheritances decrease savings (out of income) for younger households

WEALTH INEQUALITY: WITH VS. WITHOUT INHERITANCES

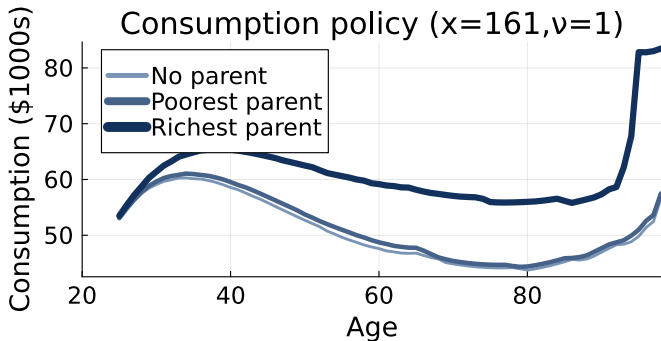


- Inheritances $\Rightarrow$ increase wealth inequality among young but lower it for old
- Partially resolves old debate on inheritances and wealth inequality (Modigliani (1988))

CONSUMPTION GRADIENT BY PARENTAL WEALTH

CONSUMPTION BY PARENTAL WEALTH

- Consumption policy functions have strong **anticipation effect**
 - Poor parent $\approx$ dead parent
 - Wealthier parents, all else equal, increases consumption by \$5-20k per year
- High parent wealth today $\Rightarrow$ higher consumption today



CONSUMPTION GRADIENT: REGRESSION

- Compare simulation to data
- Regress consumption on parental wealth quintile dummies
 - Controls: own wealth, income, and demographics.
 - Regression details
- Q1 = reference group
- Model matches PSID gradient: richer parents $\Rightarrow$ higher consumption



THE VALUE OF INHERITING EARLY

...or, why won't you die yet?

WILLINGNESS TO PAY FOR INHERITANCE RECEIPT TODAY

- How much would a household **pay** to receive the inheritance **today**?

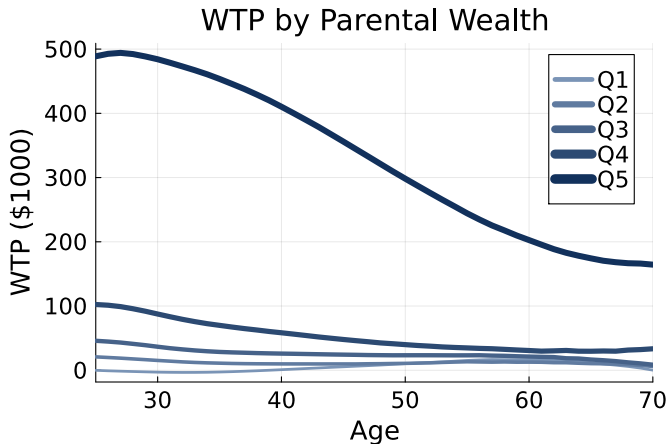
$$V(x, v, p, a) = V(x - \text{WTP} + w(1 - \tau), v, \text{dead}, a)$$

- $\text{WTP} > 0$: household benefits from early death
- $\text{WTP} < 0$: option value of living parent exceeds inheritance.

HOUSEHOLDS WOULD PAY A LOT TO INHERIT TODAY

- WTP is positive
- Decreasing in parental wealth
- Decreasing in age
 1. Need money less
 2. Shorter time to inheritance
 3. Avoid parental wealth loss

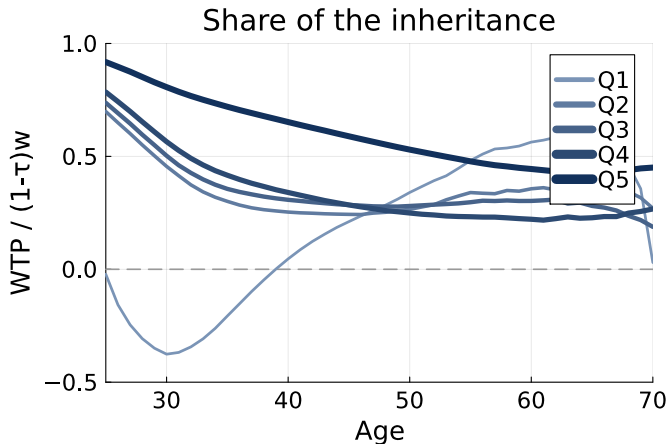
▸ Net inheritance by age



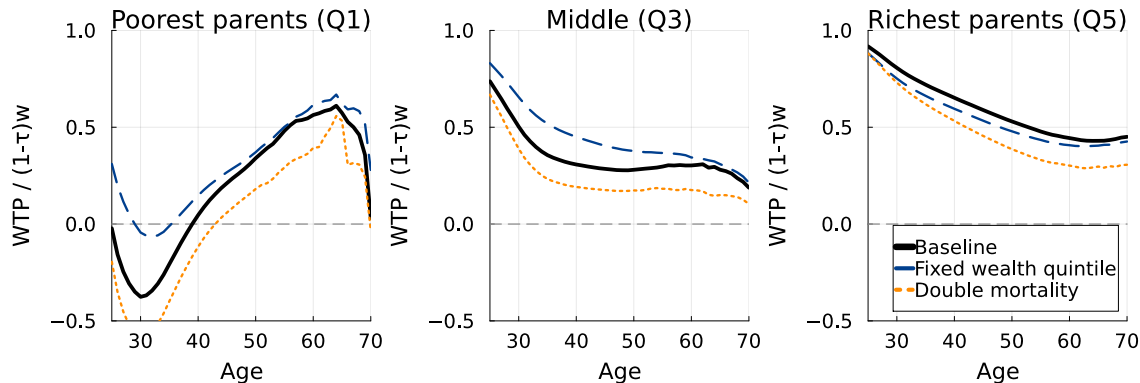
WTP NORMALIZED BY INHERITANCE

- Rich parents: highest WTP
 - Waiting is most expensive for them
- Poor parents: negative WTP
 - Keep the parent alive
- Middle (Q2–Q4): all the same

▸ Slippage ▸ Poor parents, levels



DECOMPOSING THE WTP, AS A SHARE OF THE INHERITANCE



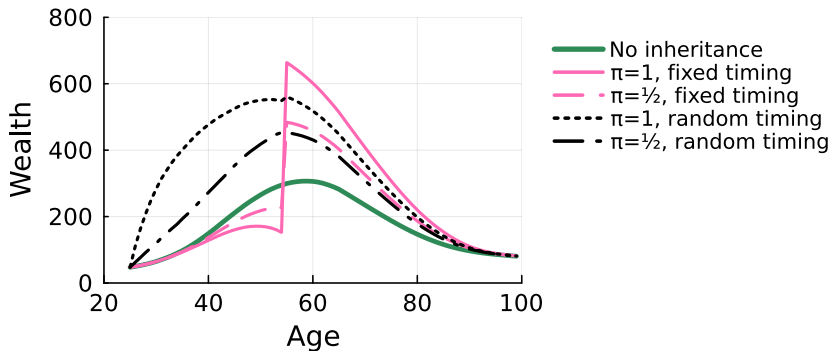
- **Fixing the quintile:** WTP rises for Q1–Q4, falls for Q5
 - Mobility is upside for everyone but the top — Q5 can only fall
- **Doubling mortality:** WTP falls — the transfer is coming anyway

CONCLUSION

- Background: Model omits inheritances, source of 25-50% of private wealth
- Question: Are inheritances important?
- Method: Extend life-cycle model to include inheritances
- Results:
 - Calibrated Inheritances $\Rightarrow$ large positive income shocks
 - Omitting inheritances $\Rightarrow$ overestimate discount factor β
 - Inheritances increase consumption more than wealth
- Next steps:
 - Event study of wealth and consumption after inheritances
 - Are inheritance expectations well calibrated?

UNCERTAIN TIMING OF INHERITANCE

Adding **random timing** (black): inheritance may arrive at any age before 55. **Known timing** is pink; certain receipt is solid, 50/50 receipt is dashed.



STANDARD LIFE-CYCLE MODEL

- Households live from age 25 to 100, retire at 65
- Choose consumption c to maximize lifetime utility:

$$V(x, v, a) = \max_c u(c) + \beta p_a \mathbb{E}[V(x', v', a + 1)] + \beta(1 - p_a) \phi(x')$$

- CRRA flow utility $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$, warm-glow bequest $\phi(x) = \psi_0 \frac{(x+\psi_1)^{1-\gamma}}{1-\gamma}$
- Budget constraint:

$$x' = (1 + r)(x - c + y(v, \varepsilon))$$

- Income: Deterministic hump shape + persistent (v) + transitory (ε) shocks,

ADDING INHERITANCES

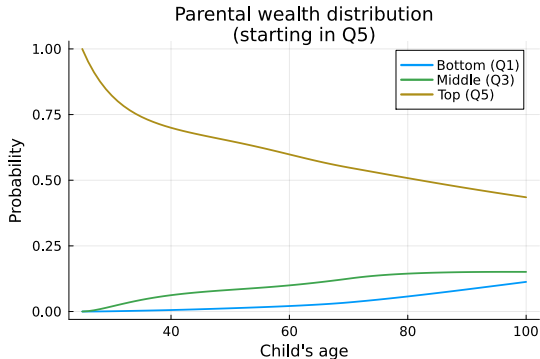
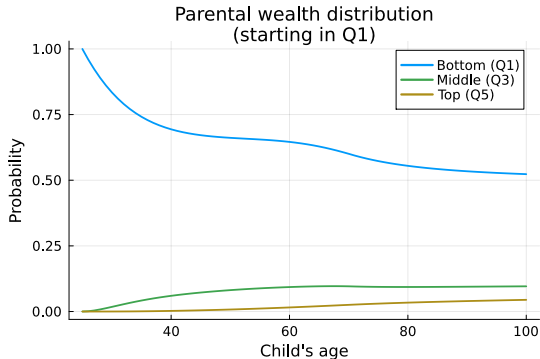
- Parental status $p \in \{\text{dead}, w_1, \dots, w_5\}$
- Extended value function:

$$V(x, v, p, a) = \max_c u(c) + \beta p_a \mathbb{E}[V(x', v', p', a + 1)] + \beta(1 - p_a) \phi(x')$$

- Budget constraint: $x' = (1 + r)(x - c + y(v, \varepsilon) + \mathbf{1}_{\text{death}} \cdot (1 - \tau(w)) w)$
- Parental transitions:
 - Survival: age-varying SSA death probabilities
 - Wealth: quintile-to-quintile transitions from PSID
 - Slippage $\tau(w)$: taxes, fees, other heirs (varies by quintile)
- Once parent dies ($p = \text{dead}$), model collapses to the standard life-cycle problem

PARENTAL WEALTH QUINTILE DYNAMICS

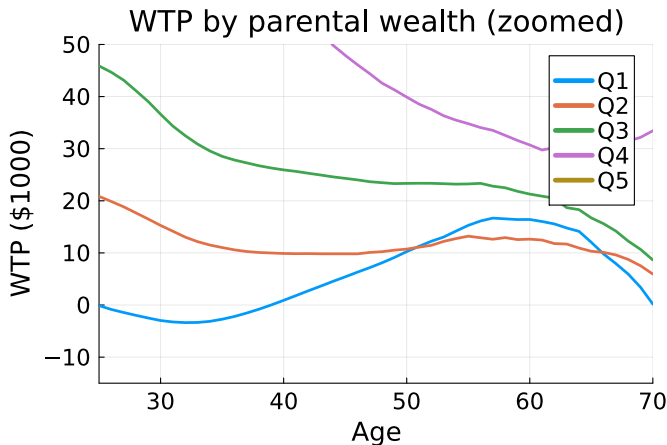
Probability of being in each quintile by age, given initial parental wealth at age 25.



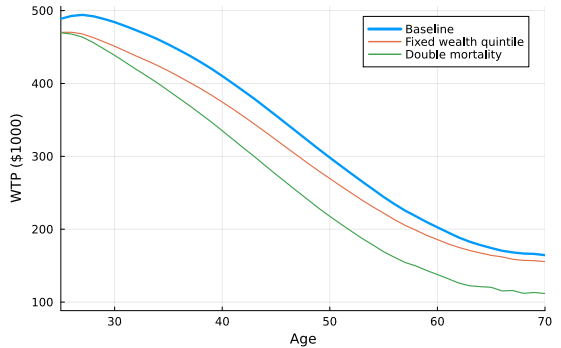
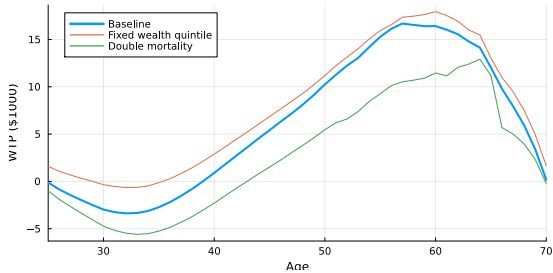
CHILDREN OF YOUNG POOR PARENTS HAVE OPTION VALUE

- **Negative WTP** until 40
 - Keep parent alive in case they get rich
- Positive wtp after 40
 - Even poor parents have some wealth to leave

▸ Net inheritance by age ▸ Return



WTP DECOMPOSITION BY PARENTAL WEALTH



Left: poorest parents. Right: richest parents.

► Return

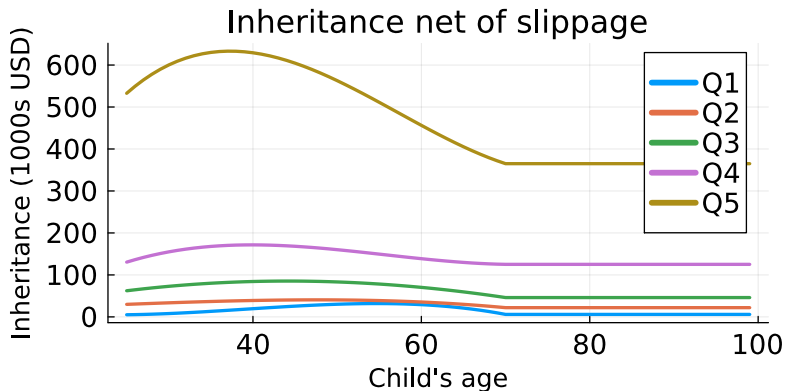
REGRESSION SPECIFICATION

$$c_{it} = \alpha + \sum_{q=2}^5 \beta_q \mathbf{1}[Q_{it} = q] + \gamma' X_{it} + \varepsilon_{it}$$

Q_{it} : parental wealth quintile (Q1 = reference)

- **PSID:** X_{it} = own wealth, quadratic in income, quadratic in age, education, race, marital status, homeownership, survey year. Family weights, clustered SEs.
- **Model:** X_{it} = own wealth, income, persistent income state indicators, quadratic in age.

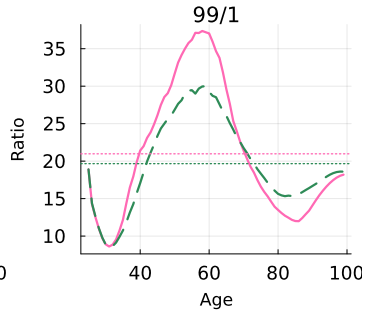
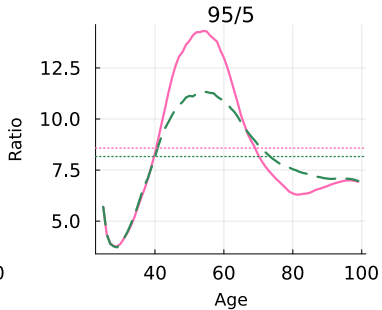
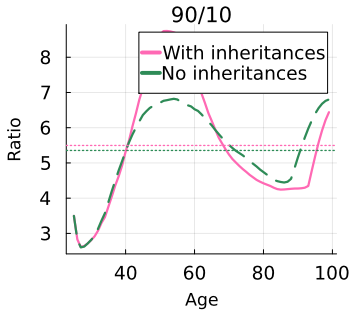
INHERITANCE NET OF SLIPPAGE BY CHILD'S AGE



Amount received $(1 - \tau(w)) w(a)$ if parent dies at each age, by parental wealth quintile.

▸ Slippage ▸ WTP

WEALTH INEQUALITY – SPREAD



▸ Return